

Scratch: Draw Shapes with Pen

Learn to create colorful geometric shapes and creative artwork using code!

1. Introduction to Scratch

Scratch is an interactive visual programming language that provides a fun way to tell stories, build games, and draw using code. Designed especially for young learners, Scratch helps you understand foundational coding logic through simple, easy-to-use drag-and-drop coding blocks. By snapping blocks together, you can create your own projects and bring characters to life.

2. Meet the Pen Extension

The **Pen Extension** allows your character (called a sprite) to act exactly like a real pen or marker on the screen. As the sprite moves according to your code, it carries the pen along and leaves a solid trail behind it. You can programmatically control when to draw, pick up the pen, adjust the color, or alter line thickness.

Core Pen Capabilities:

- **Pen Down:** Starts drawing by lowering the pen onto the canvas. It will leave a trail wherever the sprite goes.
- **Pen Up:** Stops drawing by lifting the pen. This allows the sprite to jump to a new spot without drawing a line.
- **Set Pen Color:** Pick any color from the palette to style your drawing.
- **Set Pen Size:** Change the thickness of the line to be thin or thick.
- **Clear:** Completely erases all drawings off the screen to start fresh.
- **Stamp:** Leaves a permanent visual copy of the sprite's current image on the background canvas.

3. How to Add the Pen Extension

By default, the Pen blocks are hidden. Follow these simple steps to activate them in your project palette:

- 1 Open your Scratch project editor.
- 2 Click the blue **Extensions** button located at the very bottom-left corner of the screen.
- 3 Look through the extension list to find and click on the green **Pen** card.
- 4 A brand new green category of Pen blocks will instantly appear in your blocks palette!

4. Drawing Basic Lines & Shapes

Drawing a Straight Line

To begin a drawing, you must clear old lines, place the pen down, and move the sprite across the screen.

```
when green flag clicked
  clear
  pen down // Drawing ON: leave a trail wherever I go
  move 100 steps
```

Drawing a Square

A square consists of 4 equal sides and 4 corner turns of exactly 90° . Instead of writing the code out four separate times, we use a loop block to automate the process seamlessly.

```
pen down
repeat 4 times:
  move 100 steps
  turn right 90 degrees
```

Drawing a Triangle

To draw a perfect equilateral triangle, you need 3 sides. Each turn must be 120° (not 60°) because the sprite needs to turn the *outside angle* to change direction correctly.

```
pen down
repeat 3 times:
  move 100 steps
  turn right 120 degrees
```

Drawing a Rectangle

A rectangle has 2 matching long sides and 2 matching short sides. We repeat a sequence of drawing a long side, turning, drawing a short side, and turning.

```
pen down
repeat 2 times:
  move 150 steps // Long side
  turn right 90 degrees
  move 80 steps // Short side
  turn right 90 degrees
```

Drawing a Circle

The secret trick to rendering a smooth circle is combining microscopic steps with tiny turns over and over until you complete a full rotation ($36 \times 10^\circ = 360^\circ$).

```
pen down
repeat 36 times:
  move 10 steps
  turn right 10 degrees // Small moves + small turns = smooth curve
```

5. Adding Color, Size, and Patterns

Make your digital drawings pop by modifying the appearance of the ink trail. You can select specific colors or shift them continuously inside a loop to construct a vibrant rainbow design.

- **Set Pen Color:** Locks in an exact starting color.
- **Change Pen Color by (value):** Dynamically shifts colors through the rainbow color spectrum.
- **Set Pen Size:** Establishes how bold your brush outline looks.
- **Change Pen Size by (value):** Expands or shrinks your outline thickness dynamically as you draw.

Creative Art Coding Tips:

- **Spirals:** Draw spirals by using a variable to increase the move steps slightly on every single iteration of a loop.
- **Geometric Stars:** Construct unique stars by programming sharp, narrow turn angles inside your repeat blocks.
- **Flower Petals:** Chain multiple loop blocks together to rotate a shape repeatedly around a central pivot point.

6. Hands-On Coding Challenges

Put your knowledge to the test! Try modifying your code blocks to accomplish these artistic challenges:

- 1 **Build a House:** Assemble a square base and put a triangle on top as the roof. Remember to lift the pen up when shifting positions!
- 2 **Rainbow Flower:** Create a colorful pattern out of repeating loops while changing pen colors.
- 3 **Bright Star:** Experiment with repeat loops and custom pen block configurations to construct a clean star.
- 4 **Brush Master:** Write code that alters pen size dynamically as it sweeps across the stage.

Great job today! Keep practicing, exploring, and building!